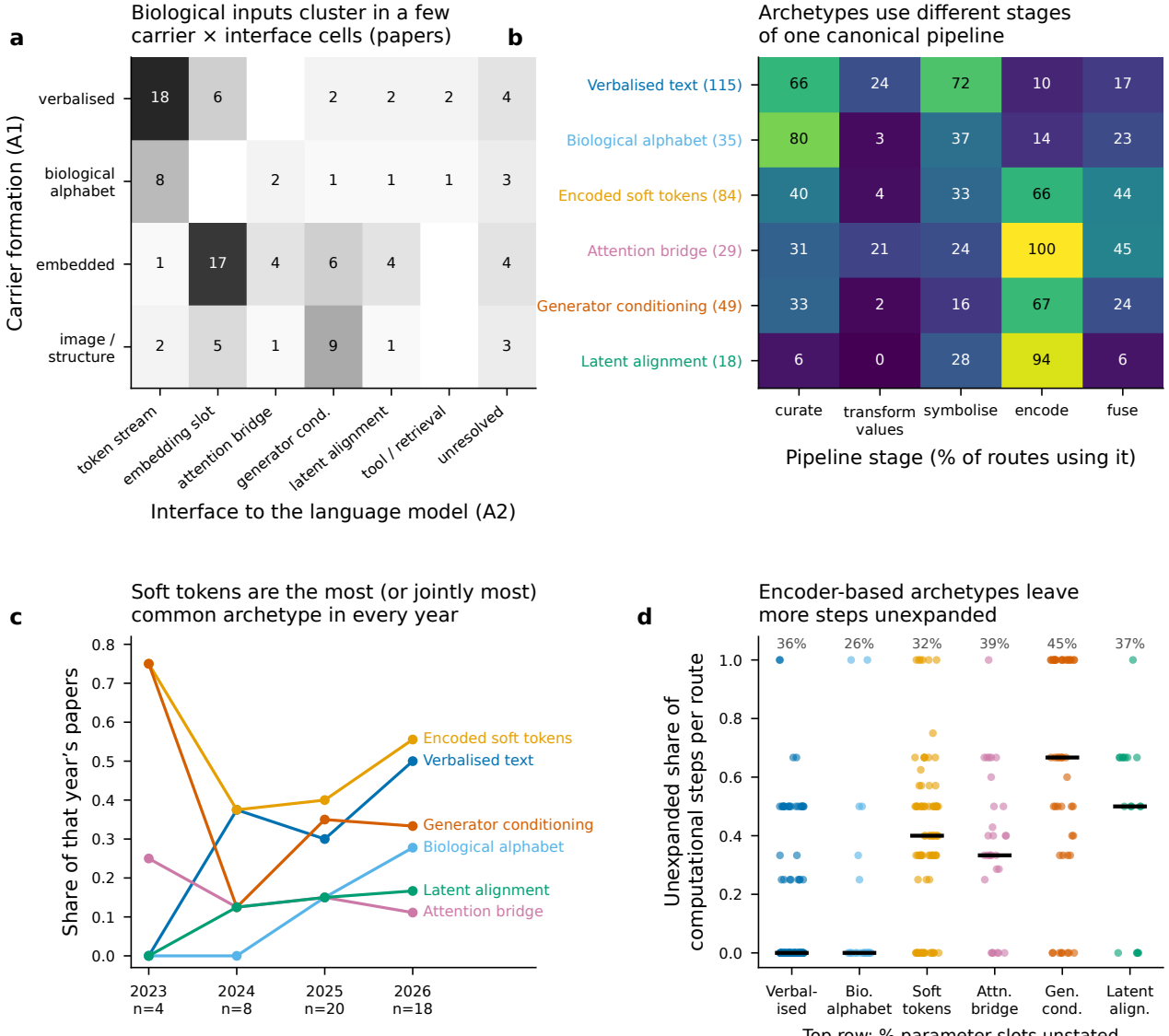

a
Biological inputs cluster in a few carrier × interface cells (papers)
Carrier formation (A1)
verbalised
biological alphabet
embedded
image / structure
18 6 2 2 2 4
8 2 1 1 1 3
1 17 4 6 4 4
2 5 1 9 1 3
token stream
embedding slot
attention bridge
generator cond.
latent alignment
tool / retrieval
unresolved
Interface to the language model (A2)
b
Archetypes use different stages of one canonical pipeline
Verbalised text (115) 66 24 72 10 17
Biological alphabet (35) 80 3 37 14 23
Encoded soft tokens (84) 40 4 33 66 44
Attention bridge (29) 31 21 24 100 45
Generator conditioning (49) 33 2 16 67 24
Latent alignment (18) 6 0 28 94 6
curate
transform values
symbolise
encode
fuse
Pipeline stage (% of routes using it)
c
Soft tokens are the most (or jointly most) common archetype in every year
Share of that year's papers
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8
Encoded soft tokens
Verbalised text
Generator conditioning
Biological alphabet
Latent alignment
Attention bridge
2023 n=4
2024 n=8
2025 n=20
2026 n=18
d
Encoder-based archetypes leave more steps unexpanded
Unexpanded share of computational steps per route
0.0 0.2 0.4 0.6 0.8 1.0
36% 26% 32% 39% 45% 37%
Verbal-ised
Bio. alphabet
Soft tokens
Attn. bridge
Gen. cond.
Latent align.
Top row: % parameter slots unstated